

Initial Thoughts for Privacy MVP:

Here are my thoughts on setting up a process for establishing the initial buildout for the Taxonomies and Ontology so we can support the following Privacy MVP outline:

Taxonomies (initial build)

Define Concept Hierarchies (class models) that can be Mapped To: *

- Applications (Vast)
- UPIA (VGRC)
- Decision Logic (Scenarios Triggering UPIA Activities)
- Should be extendable to artifacts pulled in from the Catalog
 - Metamodel for Data Domains
 - Custom Attributes
 - Custom Assets

Ontology (initial build with Change / Release Management Process)

- Draft RDF (triples) for VAST
- Draft RDF (triples) for VGRC
- Draft RDF (triples) for Catalog Artifacts
- Draft RDF (triples) for Decision Logic**

***These Taxonomies**

Knowledge Ingestion:

- Create RDF (triples) for each of the above
- Load into Knowledge Graph
- Create Relationships between Nodes (entities) in Triples and approved Terms in Taxonomy
 - Business Terms
 - Policy Terms
 - Semantics (extensions)

****Decision Logic is derived from:**

Incorporate Scenarios from Consumer Needs

- Persona + Jobs
 - Desired Outcomes
 - Feature Breakdown (FF & NFF)
 - Scenarios for triggering Discrete Activities

Note:

- Initially, we only need to map the scenarios into the Ontology
- Eventually, we should consider mapping the entire lineage for Requirements thru Scenarios to drive BEHAVIORS of Services (automated and/or manual)***

***Emergent Synthetic Services towards autonomous computing

Knowledge Engineering Design Standards:

Taxonomies are used for:

- Navigation
- Classification
- Integration

Create a “Balancing Act” between:

- Absolute granularity
 - Ultimate classification
- AND**
- Usability
 - Findability
 - Simplicity

Good Design for Taxonomies should:

- Provide structure to unstructured information
- Pull together disparately located content
- Allow consumers to discover and learn
- Extend Search with consistent POV meaning

Categorization Schemes we may want to consider:

- Facet-based – info is categorized into multiple taxonomies (stacks) based on pervasive characteristics (topic, function, ..)
- Subject-oriented – categorize based on a subject or topic
- Functional
- Organizational
- Information Type
- We can leverage a Facet-based approach to accommodate all of these POV to provide max flexibility

Taxonomy Design:

- **Start** with a Manual / Top-Down
 - Librarian science, Functional / SMEs

- Identify overall ontology and major categories of info
- Subdivide categories as necessary
- Individual-driven (can be biased)
- **Improve** using Automated / Bottom-Up Learning
 - Identify overall ontology and major collections
 - Analyze content collections using automated text analysis tools
 - Reveal major & minor topics, build taxonomy based on relationships between topics
 - Reveal new associations

Avoid Complexity in Design:

- Not too deep or wide
- No jargon or acronyms
- Limited number of taxonomies
- Limit the number of terms in any one taxonomy
- Adopt “Dot” notation IFF we need to instantiate multiple POV in a term
- Limit the number of instantiated terms (e.g. Date.Started, Date.Created, Date.Closed)
- Minimize variability

Start with your Use Cases (Consumer Needs):

1. How will your consumers be using taxonomies?
 - What are the jobs they are trying to get done?
 - How many ways do they need to think about & find info?
 - What are the most common uses of taxonomies?
2. Define the multiple sets of taxonomies consumers may want to use?
3. Use familiar terms and organize them in intuitive schemas (structures)?
 - Familiar browsing and search patterns
4. Understand the variable that impact meaning & understanding
 - New employee vs experienced SME
 - Technologist vs business person
 - Older vs younger
 - US vs International (Europe, Asia, South America, Australia, India)
 - Language variances
 - Part-time vs full time
 - Executive vs Operational staff
 - Compliance staff vs other staff
5. How do the providers need to maintain taxonomies?
 - Level of complexity required
 - Acceptable effort to maintain relationships between data, info, and content with taxonomies

- Number of relationships vs fields (attributes, tags, characteristics) to maintain
- Flexibility and adaptability of meaning as taxonomies evolve

6. Taxonomy Discipline:

- Roles & Responsibilities
 - Context Owners vs Content Owners
 - Context Review to address Consumer & Partner needs
- Policies
 - Naming conventions
 - Relationships vs. property labels (tags)
- Processes
 - Lifecycle Management
 - Change Management
 - Release Management
 - Operations Management
 - Measurement
- Design Principles:
 - Flatter is Better (easier to search)
 - If a term requires multiple qualifiers, then it's a polymorphic instance of meaning (related to one or more taxonomies)
 - Simpler (easier to understand)
 - 2-4 levels max
 - Control Breadth (should be clear / obvious)
 - Ideally, 10-15 top level Concepts
 - Continuous Learning (iteration & adaptation)
 - Learning & lineage requires Measurement
 - Validate and Publish with care
 - Invest in dedicated resources
 - Establish a strong foundation
 - Allow for extensibility
 - Consumer Driven (diversity is key)
 - Measure value once consumers & partners begin Using the taxonomies
 - Navigation
 - Classification
 - Integration
 - Measure IMPACT of change
 - Planned & Unplanned consequences

Initial Taxonomy Footprint - Getting Started:

Start with an initial list of Topics & back into the upper classification of Concepts

- People
 - Customers
 - Vendors
 - Employees
 -
 - Products
 - Services
 - Business
 - Operational
 - Technical
 - Compliance
 - Privacy
 - Security
 - Data Access
 - Releasability
 - sGDM
 - GSAM
 - Functions
 - Finance
 - Marketing
 - Retail Management
 - Sales
 - Legal
 - Audit
 - Technology
 - Risk Management
 - Human Resources
 - Facilities Management
 - Network Operations
7. Implement, Learn, Refine, Refactor
- Tie back to consumer needs
 - Rinse and Repeat